

Bootcamp **Web Development** **Fundamentals**

Delegate **Exercise Guide**

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1. Quick Labs Guide | HTML

1.1. Quick labs Environment Set-up

- 1) Open **VSCode**.
- 2) Using **File --> Open**, navigate to the **Quick Labs** folder and click **Open**. This will give you access to all of the Quick Lab files and solutions needed to complete the Quick Labs.
- 3) At the bottom of the **VSCode** window, you should see a **Go Live** icon. Click on this will launch **live-server** for the currently selected file or folder. If a folder is selected, this will be shown in your browser, and you will be able to navigate here. If a file is selected, the browser will attempt to display the file.

1.2. Quick Lab 1 — Structural HTML

Objectives

- Use structural elements appropriately

Activity

- 1) Open the file **index.html** from the **QuickLabs/01_StructuralHTML/starter** folder using **live-server** (please see step 3 of setting up the environment for help on this).
- 2) Replace any element with an appropriate structural element.

Once you have finished, check your outline matches the one below by testing the HTML structure on the website:

<https://gsnedders.html5.org/outliner/>

```
1. My Blog Posts
  1. Latest Post
    1. My HTML5 Blog
      1. HTML5 Musings
```

This is the end of Quick Lab 1

1.3. Quick Lab 2 — Hyperlinks

Objectives

- Use hyperlinks to create links between pages and actions

Activity

- 1) Open the file **index.html** from the **QuickLabs/02_Hyperlinks/starter** folder using live-server (please see step 3 of setting up the environment for help on this).
- 2) Create a new file in the same folder called **courses.html**.
- 3) In this file, skeleton an HTML page and then insert the text provided in **courses.txt** into the body.
- 4) Create a link from **index.html** to **courses.html** and vice versa. Check that these links work.
- 5) Add a link to send an email to the webmaster at the bottom of **index.html** and put this footer on the **courses** page too.

This is the end of Quick Lab 2

1.4. Quick Lab 3 — Images

Objectives

- Insert images into pages

Activity

- 1) Open the file **team.html** from the **QuickLabs/03_Images/starter** folder using **live-server** (please see step 3 of setting up the environment for help on this).
- 2) Locate the comments and insert the image in place of the comment.
- 3) The image **theTeam** causes the horizontal scroll bar to appear. Make this disappear by adding an appropriate attribute to the image.

This is the end of Quick Lab 3

1.5. Quick Lab 4 — Tables

Objectives

- Create a table to contain data with appropriate tagging

Activity

- 1) Open the file **index.html** from the **QuickLabs/04_Tables/starter** folder using **live-server** (please see step 3 of setting up the environment for help on this).
- 2) First, orientate yourself with the code. You'll see the course content within the html file, but it has no markup to describe how it should be laid out — so as you can imagine, the page is a bit of a mess.
- 3) The first step is to add our **<table>** tags around the data we want tabulated.
- 4) Now, break up the content into its logical rows using the **<tr>** tag you've learnt about already. You may be able to see what data should be on each row — but if not, ask your instructor for some guidance.
- 5) Finally, you want to now split each row into its various 'columns' by wrapping each piece of data in **<td>** tags.

So, we've gone from an illegible string of data to a highly structured, coherent table. For a bit of polish, let's tidy our code up a bit.

- 6) Without headings, our table is ambiguous. So, before the first row, add in another row and create a header for each column using the **<th>** element.
- 7) It is best practice to wrap your table headers in a **<thead>** element. It not only provides some syntactic sugar but also enables things such as scrolling table bodies separately from the headers, and when printing long tables that may span several pages the header information can be repeated on the page break. Add this element.
- 8) Add **<tbody>** tags around the table content.
- 9) Add a footer to the table with some content that indicates that all prices are inclusive of VAT.
- 10) Finally, we would rather our heading 'A sample of our courses' was tied to the data it is actually representing. For this we use the **<caption>** element which needs to be the first child of the **<table>** element. Your page should look something like this:



Learn. To Change.



A sample of our course

Course Title	Subject Area	Length	Cost
AWS Technical Essentials	Technical IT	1 day	£495
Configuring and Deploying a Private Cloud	Technical IT	5 days	£2066
PRINCE2 Foundation	Project Management	2 days	£898
Brilliant Virtual Delivery	Leadership & Management	2 days	£1362

All prices inclusive of VAT

This is the end of Quick Lab 4

1.6. Quick Lab 5 — Forms

Objectives

- To be able to create a form with several different input types

Activity

- 1) Open the file **registration.html** from the **QuickLabs/05_Forms/starter** folder using **live-server** (please see step 3 of setting up the environment for help on this).
- 2) Create a **form** where you currently see the `<!-- insert form here-->`
- 3) Create the first **fieldset** and give it a legend with **"Your Details"** as the text
- 4) Using `<label>` and `<input type="text">` elements, add user input fields for:
 - a. First name
 - b. Surname
 - c. Position / Role
- 5) There are some pre-made styles for us to utilise. Wrap everything within this **fieldset** in an **unordered list** and wrap each label/input pair in a **list item**. You'll see the styling impact that has on our form.
- 6) Add the attribute **autofocus** to the 'first name' field, save the file, and test it in your browser.
- 7) Create a new **fieldset** at the bottom of the form with a submit button.
- 8) Inside the first **fieldset** create a new list item and within it add a label/input pair with a **name**, **id**, and **type** of **"email"**.
- 9) Add a **placeholder** attribute with a value of **"you@domain.com"** and a **required** attribute.
- 10) Repeat once more adding an **input** of type **tel** with a **name** and **id** of **phone**, making it mandatory and add a placeholder with a value of **"E.g. +447500000000"**.
- 11) Create a new **fieldset** for **"Course Details"** before the **fieldset** which contains the **submit** button.
- 12) Using the same list structure in the previous **fieldset**, create a **select** element which has the following **options**:
 - a. Programming with JavaScript
 - b. Developing Web Applications using HTML5
 - c. Developing Responsive Websites
 - d. Leveraging the Power of jQuery
- 13) Add a suitable input for **"Number of delegates"**

14) Finally, create an input with an id, name, and type of "date"

15) Test your form in multiple browsers. You may find that some elements render slightly differently in different browsers — you should find a date picker UI in those browsers that support the date input type, and a text field in those that don't.

This is the end of Quick Lab 5

2. Quick Labs Guide | CSS

2.1. Quick labs Environment Set-up

- 1) Open **VSCode**.
- 2) Using **File --> Open**, navigate to the **Quick Labs** folder and click **Open**. This will give you access to all of the Quick Lab files and solutions needed to complete the Quick Labs.
- 3) At the bottom of the **VSCode** window, you should see a **Go Live** icon. Click on this will launch **live-server** for the currently selected file or folder. If a folder is selected, this will be shown in your browser, and you will be able to navigate here. If a file is selected, the browser will attempt to display the file.

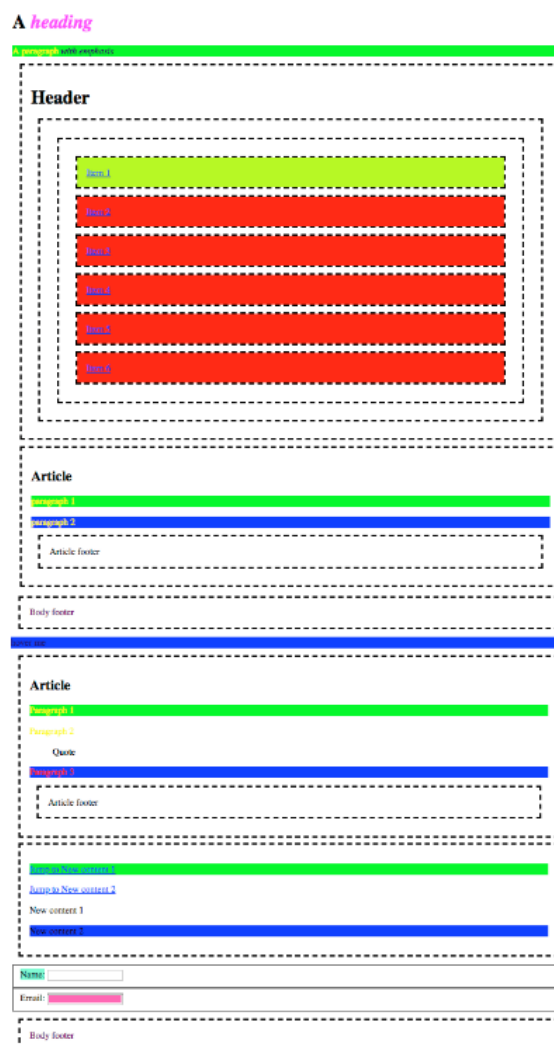
2.2. Quick Lab 6 — CSS Selectors

Objectives

- Use CSS selectors to apply styling to particular elements

Activity

- 1) Open the file index.html from the QuickLabs/06_Selectors/starter folder using live-server (please see step 3 of setting up the environment for help on this).
- 2) Apply an appropriate selector for the element(s) specified in the comment to the styling rule before it on each line.
- 3) When you have completed the first set of list-item styling check the output.
- 4) Your finished product should look like the image below. Inspect elements to find out what happened to styles that are not shown...



This is the end of Quick Lab 6

2.3. Quick Lab 7 — Text, Colours and Sizing

Objectives

- Experiment with adding colours and sizing to text and other elements

Activity

- 1) Open the file index.html from the **QuickLabs/07_TextColourSizing/starter** folder using **live-server** (please see step 3 of setting up the environment for help on this).
- 2) Apply these styling rules to the text:

Element	Styles
h1 inside a header	• A shadow that is hotpink, set 5px horizontal and 3px vertical • Text colour set to darkblue • A font family of Helvetica • A background colour set to grey of 200, 200, 200, 1 on the rgbascale • A font size of point 48 • Centre the display of the text
H1	• A font family of Arial • A font size set to 2 x the relative (em)
A p with a class of first	• Indent the text by 30 pixels • Set the colour to be red with half transparency Make the text fill the row (i.e., justify)
H2 and h3	• Align the text to the right • Use the Courier New font • Have a background of a mid-grey with half transparency
A p with an id of highlightme	• Set the background with a Hue of 45, a Saturation of 100% and a Lightness of 50%. It should be half transparent too • Space the lines by 30px • Occupy 50% of the available width
A p directly inside a footer	• A background colour of hotpink (set as a HSLA value) • Use any sans-serif font • Centre the text • Make the text uppercase

Your finished product should look like the image below:

Lorem ipsum dolor sit.

Lorem ipsum dolor sit amet.

Lorem ipsum dolor sit amet consectetur adipisicing elit. Eum, quod sequi numquam porro dolorum quos aut dolorem quidem tempora veniam, vero in sit. Accusantium dolorem minima nemo impedit deserunt, iusto aperiam sint, adipisci, at numquam cupiditate commodi blanditiis quae. Qui ut repellendus facilis sunt perspiciatis consequuntur quod, ex nulla odit harum voluptates consectetur numquam ducimus animi hic neque reiciendis! Nemo dolorum voluptate beatae, molestias quod repudiandae maxime veritatis, hic vitae consequuntur velit. Nihil, cum architecto assumenda possimus harum hic nisi distinctio necessitatibus quam, doloremque vero sunt autem quod! Quis consequatur numquam earum, nisi saepe non incidunt repellat magnam dignissimos qui corporis, harum laborum, tempore cumque eos deserunt omnis. Harum similique earum ratione fugiat. Doloribus fuga, aliquam quod omnis cupiditate ipsum ipsam eos nulla libero, numquam non placeat sunt nam minima unde at consequuntur? Libero numquam accusamus deserunt nisi eius pariatur maiores animi ullam nostrum iste, dicta laudantium natus in ab cupiditate aliquam! Reprehenderit esse deleniti ad quisquam quod provident ut incidunt sed nihil quam labore ullam, cupiditate saepe adipisci illo dolorem, corporis, maxime optio neque natus corrupti pariatur. Blanditiis, impedit ad repellendus facere dicta aliquam nam, enim nisi doloribus iste, harum nesciunt mollitia! Minima, omnis quos laudantium veritatis voluptates impedit.

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LOREM IPSUM DOLOR SIT AMET CONSECTETUR ADIPISICING ELIT. DICTA, VOLUPTATES!

This is the end of Quick Lab 7

2.4. Quick Lab 8 — Images and Backgrounds

Objectives

- Format images on pages
- Use CSS to add background images to elements on pages

Activity

- 1) Open the file **team.html** from the **QuickLabs/08_ImagesBackgrounds/starter** folder using **live-server** (please see step 3 of setting up the environment for help on this).
- 2) Add a CSS rule that makes the image occupy 50% of the available space.
- 3) Apply an image background to the **div** with an **id** of **hasBackground**. It should
 - a. Use the image named **pexels-photo-370799.jpg**.
- 4) Experiment with the **repeat** and **position** attributes.

This is the end of Quick Lab 8

2.5. Quick Lab 9 — Positioning Elements

Objectives

- Use floats, clears and other positioning and sizing properties to lay out a page

Activity

- Open the file `index.html` from the **QuickLabs/09_PositioningElements/starter** folder using **live server** (please see step 3 of setting up the environment for help on this).
- Turn the left-hand image into the right-hand image:

Positioning Elements

[Link](#) [Link](#) [Link](#)

Column

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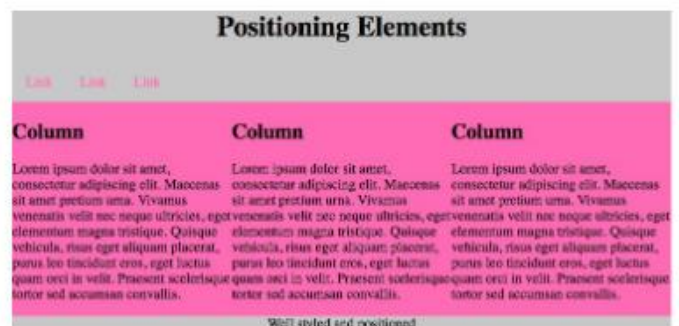
Column

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Column

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Well styled and positioned



- Guidance on CSS rules to create:

Element or Group	Items to set	Values to use
Elements of type header and footer	background-colour text-align clear	rgba(200, 200, 200, 1) center both
Elements of type nav	width height	100% 46px
Links in a nav	float display color text-align padding text-decoration	Left block hsla(300, 100%, 71%, 1) center 14px, 16px none

Hovered links in a nav	background-color color	hsla(300, 100%, 71%, 1) black
Element with id of row	clear	both
Elements with class of column	float width background-color	left 33.33% hsla(300, 100%, 71%, 1)

This is the end of Quick Lab 9

2.6. Quick Lab 10 — Tables

Objectives

- To be able to format a table using CSS

Activity

- Open the file **index.html** from the **QuickLabs/10_CSSTables/starter** folder using **live-server** (please see step 3 of setting up the environment for help on this).
- Format the table so that it looks like the image below:

Name	Country	Date of Birth
Loretta Silva	Anguilla	8/11/1985
Joel Montgomery	Nigeria	1/1/1966
Joel Brady	South Georgia & South Sandwich Islands	11/29/1960
Brent Kim	Sudan	7/21/1971
Marie Abbott	Ecuador	10/27/1967
Alvin Carter	Åland Islands	1/8/1977
Hilda Carter	Colombia	8/18/1959
Jason Cohen	Ecuador	5/12/1953
Clara Jensen	South Sudan	2/2/1986
Hallie Fields	Jordan	3/24/1960

- Ensure that the table has a hover property that highlights each row as the mouse pointer is over it as shown:

Name	Country	Date of Birth
Loretta Silva	Anguilla	8/11/1985
Joel Montgomery	Nigeria	1/1/1966
Joel Brady	South Georgia & South Sandwich Islands	11/29/1960
Brent Kim	Sudan	7/21/1971
Marie Abbott	Ecuador	10/27/1967
Alvin Carter	Åland Islands	1/8/1977
Hilda Carter	Colombia	8/18/1959
Jason Cohen	Ecuador	5/12/1953
Clara Jensen	South Sudan	2/2/1986
Hallie Fields	Jordan	3/24/1960

- The following information may help:

Element or Group	Items to set	Values to use
Element of type table	font-family border-collapse width	'Segoe UI', Tahoma, Geneva, Verdana, sans-serif collapse 100%
Elements of type th and tr	border text-align	1px solid hsla(0, 0%, 17%, 1) left
Elements of type th	padding background-color color	15px 10px hsla(330, 100%, 71%, 1) hsla(0, 0%, 17%, 1)
Even tr elements	background-color	hsla(0, 0%, 74%, 0.2)
Hovered tr elements	background-color	hsla(330, 100%, 71%, 0.5)

This is the end of Quick Lab 10

3. Quick Labs Guide | JavaScript

3.1. Quick Lab 11 — Live Output

Objectives

- Output two string variables using concatenation in the Dev Tools console in JS

Activity

- 1) Open a browser and then open developer tools.
- 2) Select the console tab.
- 3) Create two variables, your first and your last name.*
- 4) Concatenate these using any of the methods discussed to this point.
- 5) Output your full name in the console.

*You do not have to use your real name.

This is the end of Quick Lab 11

3.2. Quick Lab 12 — Live Calculation

Objectives

- Output the sum of two integer variables in the Dev Tools console using JS

Activity

- 1) Open a browser and then open developer tools.
- 2) Select the console tab.
- 3) Create two variables and store a number in each.
- 4) Output the sum of these two numbers in the console.

This is the end of Quick Lab 12

3.3. Quick Lab 13 — Live Arrays and Loops

Objectives

- Output each element of an array in the Dev Tools console using a loop in JS

Activity

- 1) Open a browser and then open developer tools.
- 2) Select the console tab.
- 3) Create an Array of your choice (e.g., fruit, cars, sports, etc.). It must have at least 4 elements in it.
- 4) Use a FOR loop to output each of the elements in your array.

This is the end of Quick Lab 13

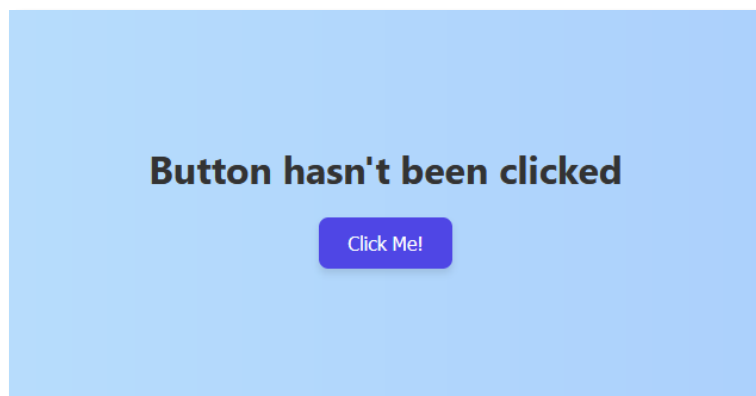
3.4. Quick Lab 14 — Buttons

Objectives

- Add JS to an HTML file to enable click event functionality of a button element

Activity

- 1) Open VS Code.
- 2) Open the file **index.html** from the **QuickLabs/14_JSButton/starter** to see the source code.
- 3) If you open this file up in a browser of your choice you will find a page with a button and some text:



- 4) Add a JS click event so when the button is pressed, the text changes to read:

"Ooo-wee! You clicked the button!"

- 5) Enter your JavaScript in the signposted section of the html file:

```
<body>

  <h1 id="status-text">Button hasn't been clicked</h1>
  <button id="click-button">Click Me!</button>

  <script>
    | // Enter your JavaScript here
  </script>

</body>
</html>
```

Stretch: Make the button a toggle, so if the user presses it again, the text returns to read

"Button hasn't been clicked"

This is the end of Quick Lab 14

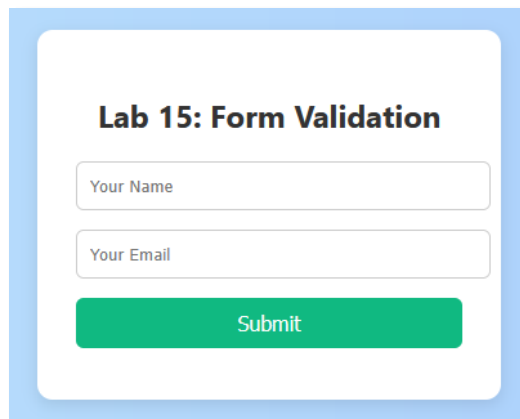
3.5. Quick Lab 15 — Form Validation

Objectives

- Add JS to an HTML file to enable client-side form validation

Activity

- 1) Open VS Code
- 2) Open the file **index.html** from the **QuickLabs/15_JSVValidation/starter** to see the source code.
- 3) If you open this file up in a browser of your choice you will find a page with a basic form and a submit button:



- 4) Add a JS click event on the submit button to validate the form fields. The only requirements of this task are to:
 - a. Validate that the two fields are not empty.
 - b. Return an error message in red if either or both of the fields are empty
 - c. If both fields are populated, return an alert that reads "Form submitted successfully!"
- 5) Enter you JavaScript in the signposted section of the starter file:

```
<div class="form-container">
  <h2>Lab 15: Form Validation</h2>
  <form id="contact-form">
    <input type="text" id="name" placeholder="Your Name" />
    <input type="email" id="email" placeholder="Your Email" />
    <button type="submit">Submit</button>
    <div id="error" class="error-message"></div>
  </form>
</div>

<script>
  // Add your form validation JavaScript here
</script>
```

Stretch: Add further JS to validate whether the email address is a valid email address and the name is ≥ 2 characters.

This is the end of Quick Lab 15

3.6. Quick Lab 16 — Image Carousel

Objectives

- Add JavaScript to an HTML file to enable the functionality of an image carousel

Activity

- 1) Open VS Code.
- 2) Open the file **index.html** from the **QuickLabs/16_JSCarousel/starter** to see the source code.
- 3) If you open this file up in a browser of your choice you will find a page with a basic carousel template. It will have a placeholder for the image, and two buttons that do not do anything. There is also an image index, though again, this is, at this stage, non-functional:



- 4) You have been provided 3 images in the **img** file, you do not have to use these images, you can source your own or use placeholders.
- 5) Add JavaScript to fulfil the following requirements:
 - a. The next and previous buttons navigate appropriately forward and backwards in the image list (Images change to the next in the index or return to the previous index).
- 6) Enter you JavaScript in the signposted section of the starter file:

```
<div class="carousel">
  
  <div class="buttons">
    <button id="prev-btn">Previous</button>
    <button id="next-btn">Next</button>
  </div>
  <div class="counter" id="image-counter">Image 1 of 3</div>
</div>

<script>
  // Add your JavaScript carousel logic here
</script>
```

Stretch: Add a picture ID under the buttons (e.g., "Picture 1 of 3") which updates when you move the picture along.

This is the end of Quick Lab 16

4. Quick Labs Guide | Responsive Design

4.1. Quick labs Environment Set-up

- 1) Open **VSCode**.
- 2) Using **File --> Open**, navigate to the **Quick Labs** folder and click **Open**. This will give you access to all of the Quick Lab files and solutions needed to complete the Quick Labs.
- 3) At the bottom of the **VSCode** window, you should see a **Go Live** icon. Click on this will launch **live-server** for the currently selected file or folder. If a folder is selected, this will be shown in your browser, and you will be able to navigate here. If a file is selected, the browser will attempt to display the file.

4.2. Quick Lab 17 — Media Queries

Objectives

- Use media queries to change the layout on screen dependent on its size

Activity

- 1) Open the files **index.html** from the **QuickLabs/17_MediaQueries/starter** folder in the browser using **live-server** (please see step 3 of setting up the environment for help on this).
- 2) In **mediaqueries.css**, add a media query that:
 - Detects if the minimum width of the screen is 794 pixels
 - Sets the **body** to have properties of:
 - background with HSLA values of 240, 100%, 50% and 0.2
 - margins top and bottom of 25px
 - margins left and right of 6%
 - Sets the **figure** to have properties of:
 - float to be left
 - width to be 40%
 - right margin to be 4.5%
- 3) Add a second media query that:
 - Sets the **body** to have properties of:
 - background with HSLA values of 120, 100%, 50% and 0.2
 - margins top and bottom of 25px
 - margins left and right of 10%
 - a maximum width of 1130px
 - Sets the **figure** to have properties of:
 - width of 28%
 - right margin to be 2.5%
- 4) Check the output in the browser - everything does not work as it should... Can you work out what is missing from the **index.html <head>**? Think VP!

This is the end of Quick Lab 17

4.3. Quick Lab 18 — Grids

Objectives

- To apply classes to HTML to utilise a Grid layout

Activity

- 1) Open the files **index.html** from the **QuickLabs/18_Grids/starter** folder in the browser using **live-server** (please see step 3 of setting up the environment for help on this).
- 2) Open **grids.css** and examine the style rules that have been placed in the file already.
- 3) You should apply the appropriate CSS classes to the elements in the HTML to make your page (when full screen) look like the image below:



For reference, the menu links are 1/4 of the screen.

- 4) Once you have succeeded, experiment using the other col-X settings.

This is the end of Quick Lab 18

4.4. Quick Lab 19 — Flexbox

Objectives

- To examine some of flexbox's properties and values

Activity

- 1) Open the files **index.html** from the **QuickLabs/19_Grids/starter** folder in the browser using **live-server** (please see step 3 of setting up the environment for help on this). Have a look at the output.
- 2) Examine the CSS classes that are applied to the 3 `<ul>` tags - you should notice that each has a **flex-container** class and then another.
- 3) In **flexbox.css** create a rule for the **flex-container** class that has:
 - **padding** and **margin** set to **0**;
 - **list-style** set to **none**;
 - A **border** that is a **solid** line of **1px** coloured **silver**;
 - **display** set to **flex**.
- 4) Create a rule for the **nowrap** class that has **flex-wrap** set to **nowrap**.
- 5) Create a rule for the **wrap** class that has **flex-wrap** set to **wrap**.
- 6) Create a rule for the **wrap-reverse** class that has **flex-wrap** set to **wrap-reverse**.

Save the file and check the output and that it responds appropriately when you shrink/expand the size of the available screen.

- 7) Create a rule for the **direction** class that has **flex-direction** set to **column**.

Again, save and check the output — the 4th set of blocks should have been affected.

- 8) Change the **flex-direction** for the **direction** class to **row**.
- 9) Add a rule for the **justify** class and set **justify-content** to **center**.

Again, save and check the output — the 4th set of blocks should have been affected.

Experiment with any of the flex properties by adding properties to the classes.

Note that some properties will not affect elements with particular others applied to them.

This is the end of Quick Lab 19

4.5. Quick Lab 20 — Responsive Images

Objectives

- To be able to use the **srcset** attribute with images.
- To use the **<source>** element and **sizes** attributes for responsive images

Activity

- 1) Open the files **index.html** from the **QuickLabs/20_MediaQueries/starter** folder in the browser using **live-server** (please see step 3 of setting up the environment for help on this).
- 2) In the first **<article>**, add an **** that:
 - Initially uses the image **"baboon1x.jpg"** as its **src**;
 - Has an **alt** of **"Baboon"**;
 - Has a **srcset** set to **"baboon1x.jpg 1x, baboon2x.jpg 2x"**.
- 3) Save the file and view in the browser - the 1x image should be shown
- 4) Use the Device Selector in the Developer Tools and **reload** the page. This should change the image to the 2x image.
- 5) In the second **<article>**, add a **<picture>**. It should have child elements of:
 - **<source>** with a **media** attribute set to a **"(min-width:1024px)"** and a **srcset** of **"kitten-large.png"**;
 - **<source>** with a **media** attribute set to a **"(min-width: 667px)"** and a **srcset** of **"kitten-medium.png"**;
 - **** with a **src** of **"kitten-small.png"** and alt of **"Cute Kitten"**.
- 6) Save the file and view in the browser. Changing the width of the device through the Device Selector being set to Responsive should change the image as the width is increased or decreased.
- 7) In the third **<article>**, add an **** with attributes set to:
 - **src** of **"small.jpg"**;
 - **alt** of **"Cute Kitten"**;
 - **sizes** set to **"50vw"**;
 - **srcset** of **"kitten-small.png 500w, kitten-medium.png 1000w, kitten-large.png 1500w"**.
- 8) Save the file and change the Responsive width of the Device Selector to be around 400 pixels.
- 9) Refresh the page and note that the **kitten-small.png** image is shown.

- 10) Increase the width of the available screen and note that the image changes to the larger ones as breakpoints are hit.
- 11) Once the **kitten-large.png** image is shown, refresh the page again.
- 12) Decrease the width of the available screen and note that the image DOES NOT change.

This is the end of Quick Lab 20

5. Capstone Project (Group)

5.1. Project Overview:

In this project, you and your peers will design and build a small, functional website using core front-end technologies: **HTML** for structure, **CSS** for styling, and **JavaScript** for interactivity. You will also need to integrate a front-end framework, such as Bootstrap, to streamline layout, responsiveness, and UI components.

Your aim is to apply skills learnt during this module by producing a website that is clear, usable, and even, potentially, responsive across different devices.

You and your peers will plan content and structure, write readable, commented code and implement at least one interactive feature using JavaScript.

5.2. Project preparation:

- This is a group project, so ensure you are in a group of no less than 2 but preferably ≥ 3
- If you are having any issues with group delegation, please ask your trainer.

5.3. Project Requirements:

- Minimum 1 page with no upper limit.
- Application of skills is better than volume of work. (**Quality over Quantity**)
- **Must include:**
 - Reasonable design documentation (a light design)
 - **Project summary (short paragraph)**
 - What is the website about?
 - Who's the target audience?
 - **Page plan / Site map**
 - **Rough content plan**
 - **Key feature list**
 - Style ideas (e.g., simple notes on colour palette or style (e.g., "clean and modern", "earth tones"))
 - **Keep it light**
 - *Suggest 1 page or 1 short document (can be handwritten or typed).*
 - *30-60 minutes max to complete.*
 - *Emphasise it's a planning tool, not a polished design spec.*

- **Skills that must be demonstrated practically:**

- HTML, CSS, and evidence of a framework (e.g., Bootstrap).
- At least 1 JavaScript element.*
- You must comment throughout to demonstrate understanding of applied approaches and techniques.

*Any element that requires submitting of data (e.g., Forms) do not have to post to anywhere. However, buttons and hyperlinks are expected to work (to internal or external content.)

- **Testing documentation (light)**

- Whereas you may not have time to test your site thoroughly, it is recommended that there is some evidence of system testing.
- Documentation, similar to the planning documentation doesn't have to be robust, but it would be good practice to have a light test plan for interactive elements on your site.

- **Presentation**

- You and your peers should be prepared to demonstrate your deliverable to your colleagues in the classroom environment.
- This doesn't have to be a full presentation, just an informal talk through of your project whilst presenting a visual to your class.
- You are required to offer a brief Q&A to give your peers a chance to ask any questions they may have on your presentation.

5.4. Deliverables:

- **Each group should be able to submit (Trainer to collect on USB stick or to set up a local shared folder):**

- Design documentation
- All HTML, CSS, JS files in a suitable folder structure (including any images or other media needed for the website to present appropriately)
- Tasting documentation (optional)
- Presentation notes / slides if applicable

5.5. Timeline:

This is **2-week project (2 days of classroom work)**. This is **NOT an enterprise project**, more a showcase of skills and approaches covered in the classroom. Please bear this in mind as subjects and project sections should be scaled to suit this **SHORT** deadline.

5.6. Tips to manage this short time window:

- Used **Lorem Ipsum** for text instead of generating text (<https://www.lipsum.com/>)
- Use **premade placeholders** instead of sourcing / making images (<https://placeholder.co/>)
- Pair back documentation: Design / Testing to a brass tacks approach (**minimalist**)
- **Focus on the skills covered in this module, specifically the practical scripting / coding and the commenting to demonstrate understanding through application**

5.7. Additional project guidance:

- Try to limit use of AI, although tempting and not forbidden, it's more valuable to include less complex elements than to include something that you don't understand.
- You won't remember everything covered, be prepared to look through content, undertake some research and most importantly, ask questions of the instructor as well as your peers!

Your trainer is there to facilitate this project. Please ask any questions or queries you may have throughout your project.



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